

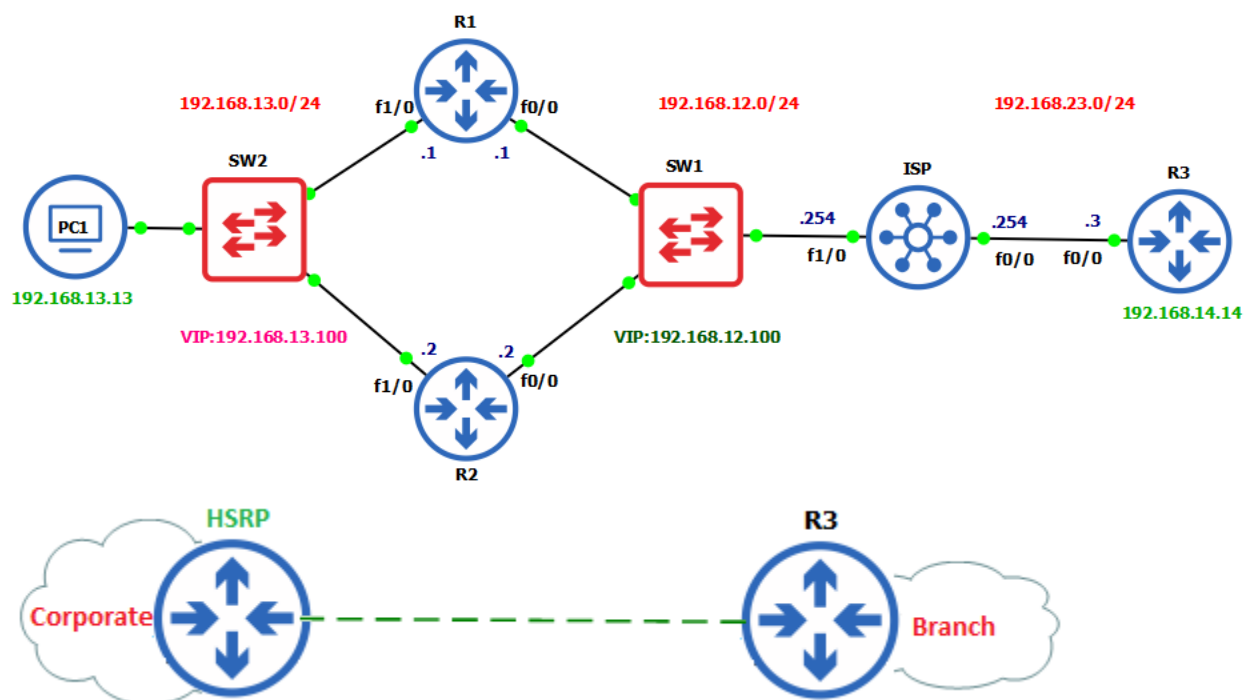
HA Consideration for Site-to-Site VPN Solutions:

HSRP:

Hot Standby Routing Protocol (HSRP) is specially designed for ensuring the high availability in the networks by using standby router which monitors the primary active router, when the primary active router goes down, it takes control hence providing redundancy. HSRP and IPsec Stateful failover enabled active router sends the following information to the standby router: IKE cookies stamp, Session keys, Cisco Service Assurance (SA) Agent attributes, Sequence number counter and window state, Kilobyte (KB) lifetime expirations and Dead peer detection (DPD) sequence number updates.

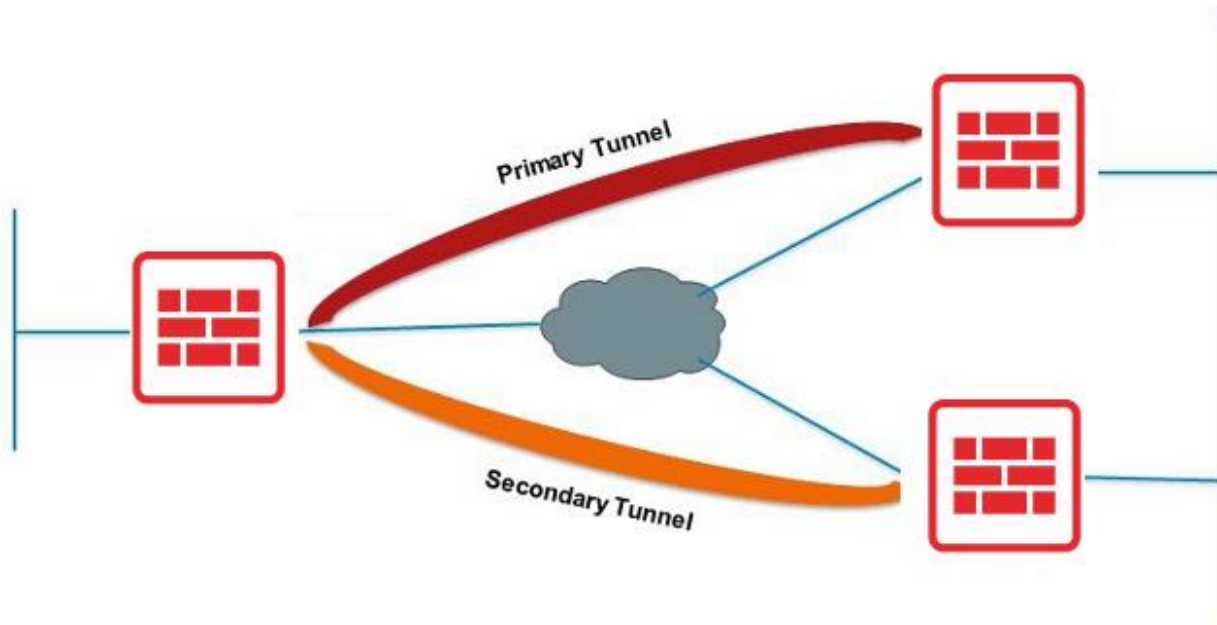
SSO:

Stateful Switchover (SSO) is an IOS feature which can provide inter-device service synchronization and stateful failover. Stateful Switchover (SSO) allows for a hot-standby processor to take control of a failed route processor while maintaining connectivity.



HA Site to Site VPN on ASA:

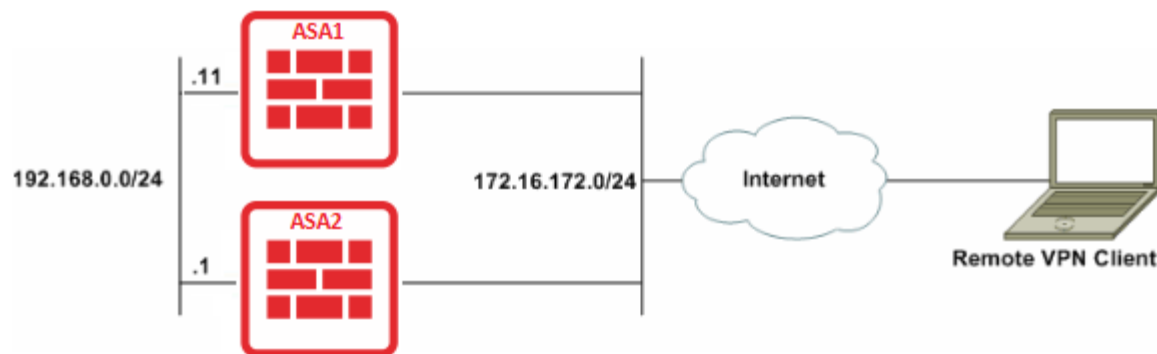
There is no preemption in IPsec site-to-site VPN on Cisco ASA to the primary peer. If you configure a crypto map with two peers, one as the primary, and another as the secondary, the ASA will try always to initiate the tunnel with the primary peer. If the primary peer fails and become unreachable, then the ASA will initiate the tunnel with the secondary peer. When the primary comes back online, the ASA will not do anything as long as the secondary is reachable. In other words, the primary peer will not take over the control from the secondary; instead the tunnel will remain initiated with the secondary peer.



HA Consideration for Remote Access VPN Solutions:

Redundant VPN Peering:

IPsec as well as AnyConnect VPN client software allow multiple ASA addresses to be configured as VPN servers. In case primary ASA is failing and becoming unavailable, before client attempts to establish a new connection, VPN software tries to connect to next available address from the list of configured addresses. Similarly, client software can use the Dead Peer Detection (DPD) feature to detect the unavailability of ASA if connection between client and ASA has already been established. As this method of redundancy is totally client-side feature, so there is no formal requirement for software version of multiple ASA being configured as VPN peers in client configuration.



VPN Clustering:

VPN clustering provides redundancy to AnyConnect users by sharing and load balancing between devices in a cluster. One device is configured to act as a master, and it will be responsible to handle the incoming connections and further distribute them to least loaded device in a cluster for further action. This method does not necessarily require each ASA device to have same software version, however, a single device with no support for single configuration command can affect the whole cluster.

